

Stream Control Transmission Protocol (SCTP)

Module 28

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Hello friends I welcome you all to the video lecture series of cyber security and information security in this lecture we will study about stream control transmission protocol that is sctp first of all please see the contents which I shall cover after covering brief introduction about sctp and sctp Services I shall take a features of sctp which will be followed by sctp packet format and last I shall take up sctp Association so now let us start stream control transmission protocol is a new reliable message oriented transport layer protocol figure shows the relationship of sctp to the other protocols TCP and UDP in the Internet Protocol suit sctp lies between the application layer and the Network layer and serves as the intermediary between the application programs and the network

operations sctp however is mostly designed for internet applications that have recently been introduced these new applications such as iua ISDN over IP m2ua and M3 UA that is telephony signaling h248 that is Media India Gateway control h.323 IP telephony N Sip IP telephony need a more sophisticated service that TCP can provide sctp provides this enhanced performance and reliability we briefly compare UDP TCP and sctp UDP is a message oriented protocol process delivers a message to UDP which is encapsulated in a user datagram and sent over the network UDP conserves the message boundaries each message is independent from any other message this is a desirable feature when we are dealing with the applications such as IP

telephony and transmission of realtime data as we will see later in the text however UDP is unreliable the sender cannot know the destination of messages sent a message can be lost duplicated or received out of order UDP also lacks some other features such as congestion control and flow control needed for a friendly transport layer protocol TCP is a byte oriented protocol it receives a message or messages from a process stores them as a stream of bytes and sends them in segments 3 minutes, 1 second there is no preservation of the message boundaries however TCP is a reliable protocol the duplicate segments are detected the LW segments are resent and the bytes are delivered to the end process in

order TCP also has congestion control and flow control mechanisms sctp combines the best features of UDP and TCP sctp is a reliable message oriented protocol it preserves the message boundaries and at the same time detects lost data duplicate data and out of order data it also has congestion control and flow control mechanisms later we will see that sctp has other Innovative features unavailable in UDP and TCP sctp is a message oriented reliable protocol that combines the best features of UDP and TCP sctp Services let us explain the services offered by sctp to the application layer processes process to process communication sctp uses all whenn ports in the TCP space following table lists some extra port numbers used by sctp protocol iua uses port number

90 protocol M2 UA uses port number 2904 protocol M3 UA users port number 2905 protocol h248 uses port number 2949 protocol h.323 users port numbers 1718 1719 1 1720 and 11720 protocol sip users port number 5060 multiple streams each connection between a TCP client and a TCP server involves one single stream the problem with this approach is that a loss at any point in the Stream blocks the delivery of the rest of the data this can be acceptable when we are transferring text it is not when we are sending realtime data such as audio or video sctp allows multistream service in each connection which is called Association in sctp terminology if one of these streams is blocked the other streams can still deliver their data the idea is similar to multiple

Lanes on a highway each lane can be used for a different type of of traffic for example one lane can be used for a regular traffic another for car pools if the traffic is blocked for regular Vehicles car pool Vehicles can still reach their destinations multi homing a TCP connection involves one source and one destination IP address this means that even if the sender or receiver is a multihomed host connected to more than one physical address with multiple IP addresses only one of these IP addresses per end can be utilized during the connection an sctp association on the other hand supports multihoming service the sending and receiving host can Define multiple IP addresses in each in for an association in this fault

tolerant approach when one path fails another interface can be used for data delivery without 7 minutes, 1 second interruption this fault tolerant feature is very helpful when we are sending and receiving a realtime playl such as Internet telepon figure shows the idea of multihoming in the figure the client is connected to two local networks with two IP addresses the server is also connected to two networks with two IP addresses the client and the server can make an association using four different pairs of IP addresses however note that in the current implementations of sctp only one pair of IP addresses can be chosen for normal communication the alternative is used if the main choice fails in other words at present sctp

does not allow load sharing between different parts satp Association allows multiple IP addresses for each end full duplex communication like TCP sctp offers full duplex service where data can flow in both directions at the same time each sctp then has a sending and receiving buffer and packets are sent in both directions connection oriented service like TCP

sctp is a connection oriented protocol how however in sctp a connection is called an association when a process at site A wants to send and receive data from another process at site B the following occurs the two processes establish an association between each other data are exchanged in both directions the association is terminated reliable service if sctp like

TCP is a reliable transport protocol it uses an acknowledgement mechanism to check the safe and sound arrival of data we will discuss this feature further in the section on error control sctp features let us discuss the general features of sctp transmission sequence number TSN the unit of data in TCP is a byte data transfer in TCP is controlled by numbering bytes using a sequence number on the other hand the unit of data in sctp is a data chunk which may or may not have a one to one relationship with the message coming from the process because of fragmentation data transfer in sctp is controlled by numbering the data chunks Sctp uses a transmission sequence number TSN to number the data chunks in other words the TSN in sctp plays the

analogous role as the sequence number in TCP tsns are 32 bits long and randomly initialized between 0 and 2³² - 1 each data chunk must carry the corresponding TSN in its header in sctp a data chunk is numbered using a transmission sequence number stream identifier si in TCP there is only one stream in each Connection in sctp there may be several streams in each Association each stream needs to be identified using a stream identifier SI each data chunk must carry the SI in its header so that when it arrives at the destination it can be properly placed in its stream the SI is a 16 bit number starting from zero to distinguish between different streams sctp uses and Si stream sequence number SSN when a data chunk arrives at the

destination sctp it is delivered to the application at stream and in the proper order this means that in addition to an SI sctp defines each data chunk in each stream with a stream sequence number SSN to distinguish between different data chunks belonging to the same stream sctp uses ssns packets in TCP a segment carries data and control information data are carried as a collection of bytes control information is defined by six control flags in the header the design of sctp is totally different data are carried as data chunks control information as control chunks several control chunks and data chunks can be packed together in a packet a packet in sctp plays the same role as a segment in TCP TCP has segments sctp has packets we will discuss the format of sctp packet

in the next section for the moment let us briefly list the differences between an sctp packet and a TCP segment the control information in TCP is part of the header the control information in Sctp is included in the control chunks there are several types of control chunks each is used for different purpose the data in TCP segment treated as one entity an sctp packet can carry several data chunks each can belong to different stream the options section which can be part of TCP segment does not exist in an sctp packet options in sctp are handled by defining new chunk types the mandatory part of TCP header is 20 bytes while the general

header in sctp is only 12 bytes the sctp header is shorter due to the

following an sctp sequence number TSN belongs to each data chunk and hence is located in the chunks header the acknowledgement number and window size are part of each control chunk there is no need for a head length field known as HL in the TCP segment because there are no options to make the length of the header variable the sctp header length is fixed 12 bytes there is no need for an urgent pointer in p as we will see later the check sum in TCP is 16 bits in sctp it is 32 bits the verification tag in sctp is an association identifier which does not exist in TCP in TCP the combination of Ip and Port addresses defines a connection in STP we may have multihoming using different IP addresses a unique verification tag is needed to Define each

Association TCP includes one sequence number in the header which defines the number of First white in the data section an actp packet can include several different data chunks TSN is and ssns Define each data chunk some segments in TCP that carry control information need to consume one sequence number control chunks in actp never use a TSN is or SSN number these three identifiers below belong only to data chunks not to the whole packet in sctp control information and data information are carried in separate chunks in STP we have data chunks streams and packets an association may send many packets a packet may contain several chunks and chunks May belong to different streams to make the definitions of these terms clear let us suppose that a process a needs to send

11 messages to process B and three streams the first four messages are in the first stream the second three messages are in the second stream and the last four messages are in the third stream although a message if long can be carried by several data chunks we assume that each message fits into one data chunk therefore we have 11 data chunks in three streams the application process delivers 11 messages to sctp where each message is ear marked for the appropriate stream although the process could deliver one message from the first stream and then another from the second we assume that it delivers all messages belonging to the first stream first all messages belonging to the second stream next and finally all messages belonging to the last stream

we also assume that the network allows only three data chunks per packet which means that we need four packets as shown in the figure data chunks in stream zero are carried in the first and part of the second packet those in stream one are carried in second and the third packet those in stream two are carried in the third and fourth packet no that each data chunk needs three identifiers TSN SI and SSN TSN is a cultive number and used as we will see later for flow control and error control s defines the stream to which the chunk belongs SSN defines the chunk order in a particular stream in our example SSN starts from zero for each stream acknowledgement number TCP acknowledgement numbers are bite

oriented and refer to the sequence numbers sctp acknowledgement numbers are chunk oriented they refer to the TSN a second difference between TCP and sctp acknowledgements is the control information recall that this information is part of the segment header in TCP to acknowledge segments that carry only control information TCP uses a sequence number and acknowledgement number for example a SYN segment needs to be acknowledged by an ACK segment in sctp however the control information is carried by the control chunks which do not need a TSN these control chunks are acknowledged by another control chunk of the appropriate type for example an init control chunk is acknowledged by an init chunk there is no need for a sequence number or an acknowledgement

number in sctp acknowledgement numbers are used to acknowledge only data chunks control chunks are acknowledged by other control chunks if necessary packet format in this section we show the format of a packet and different types of chunks most of the information presented in this section will become clear later this section can be skipped in the first reading or used only as a reference an sctp packet has a mandatory General header having 12 bytes and a set of blocks called chunks having variable length there are two types of chunks control chunks and data chunks a control chunk controls and maintains the association a data chunk carries user data 20 minutes, 1 second data in an sctp packet control chunks come before data

chunks General header the general header that is Packet header defines the end points of each Association to which the packet belongs guarantees that the packet belongs to a particular Association and preserves the Integrity of the contents of packet including the header itself there are four fields in the general header Source port address this is a 16bit field that defines the port number of the process sending the packet destination port address this is a 16bit field that defines the port number of the process receiving the packet verification tag this is a number that matches a packet to an association this PR prevents a packet from a previous Association from being mistaken as a packet in this Association it serves as an identifier for the association it is repeated in every

packet during the association there is a separate verification used for each Direction in the association checksum this 32 bit contains a crc32 checksum note that the size of the checksum is increased from 16 bits to 32 bits in sctp to allow the use of crc32 checksum chunks control information or user data are carried in chunks chunks have a common layout as shown in the figure the first three fields are common to all chunks the information field depends on the type of Chunk the important point to remember that sctp requires the information section to be a multiple of four bytes if not padding bytes are added at the end of the section chunks need to terminate on a 32bit boundary the description of the common fields are as follows type this 8bit

field can Define up to 256 types of chunks only a few have been defined so far the rest are reserved for future use see the table for list of chunks and their descriptions flag this 8bit field defines special flags that a particular chunk may need each bit has a different meaning

depending on the type of Chunk length since the size of information section is dependent on the type of Chunk we need to define the chunk boundaries the 16bit field defines the total size of the Chunk in bytes including the type flag and length Fields sctp Association sctp like TCP is a connection oriented protocol however a connection in sctp is called an association to emphasize multihoming a connection in actp is called an association association

establishment Association establishment in SC TP requires a four-way handshake in this procedure or process normally a client wants to establish an association with another process normally a server using actp as the transport layer protocol similar to TCP the actp server needs to be prepared to receive any association association establishment however is initiated by the client sctp association establishment is as shown in the figure the steps in a normal situation are as follows the client sends the first packet which contains an init chunk the verification tag VT of this packet defined in general header is zero because no verification tag has yet been defined for this direction client to

server the init tag includes an initiation tag to be used for packets from the other direction server to client the chunk also defines the initial TSN for this direction and advertises a value for RND the value of RND is normally advertised in a s chunk it is done here because sctp allows the inclusion of data Chunk in the third and fourth packets the server must be aware of the available client buffer size note that no other chunks can be sent with the first packet the server sends the second packet which contains an init acknowledgement chunk the verification tag is the value of the initial tag field in the inet chunk this chunk initiates the tag to be used in the other direction defines the

initial TSN for data flow from server to client and sets the servers rewind the value of rewind is defined to allow the client to send a data chunk with the third packet the init acknowledgement also sends a cookie that defines the state of server at this moment we will discuss the use of the cookies shortly the client sends the third packet which includes a cookie Eco chunk this is a very simple chunk that EOS without change the cookie sent by the server satp allows the included of data chunks in this packet the server sends the fourth packet which includes the cookie acknowledgement chunk that acknowledges the receipt of cookie e chunk sctp allows the inclusion of data chunks with this packet so friends this was all about

stream control transmission protocol sctp hope the concepts explained in this lecture were understandable and helpful hope to see you in the next lecture till then goodbye thank you